Shakti Kumar

STEM Subject-Matter Expert · Physics, Mathematics, Mechanical & Materials Engineering · PhD Scholar, IIT Kanpur

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Profile

PhD scholar in Materials Science & Engineering at IIT Kanpur, with a B.Tech in mechanical and an M.Tech in aerospace engineering. Writes and checks scientific code in Python (DFT with Quantum ESPRESSO, molecular dynamics with LAMMPS, phase-field and machine-learning models) in public, unit-tested repositories. Authors AI training tasks with deterministic Python graders for Alignerr, and verifies AI output against physics and published data.

Projects

Computational Materials Science Portfolio (11 projects, 60 unit tests) — github.com/Shaktikr/mse-computational-projects · 2026

- Computed aluminium's creep parameters from first principles in Quantum ESPRESSO (141 converged runs): vacancy formation 0.67 eV and migration 0.59 eV, giving Q = 1.26 eV against 1.28 eV measured.
- Reproduced Ogata, Li & Yip (*Science*, 2002) with DFT: aluminium is stronger than copper in ideal shear (3.19 vs 2.14 GPa).
- Calculated B2 NiAl elastic constants within 5% of experiment, ran seed-averaged LAMMPS creep simulations of nanocrystalline Ni, and reached 80% phase accuracy on a real 1,545-entry high-entropy-alloy dataset (58% for empirical rules).

ML Stress Prediction for Gyroid TPMS Lattices — github.com/Shaktikr/gyroid-fea-ml · course project, MSE643 (AI & ML in Materials Engineering), IIT Kanpur

- Compared five regressors on Abaqus compression data (5 gyroid geometries, 2,525 records): a feature-engineered linear model beat tree ensembles in the elastic regime (test MAE 0.14 MPa), confirmed with a bias–variance curve and grid search.

Thermal-Shock Testing of Alumina Thermal Barrier Coatings — github.com/Shaktikr/ceramic-coating-thermal-fatigue · B.Tech major project (team of five), REC Banda, 2021–22

- Thermal-shock tested eight plasma-sprayed alumina compositions above 1100 °C: only the 20 wt% 3YSZ coating survived and survival tracked cooling rate (2.10 vs 4.25 °C/s).

M.Tech Thesis: Compressible Elliptic Vortex Ring Dynamics — IIT Kanpur, 2022–24

- Shock-tube experiments with high-speed schlieren imaging; flow-field data extracted from the images in MATLAB and Python.

Experience

AI Training Task Author (freelance) — Alignerr (Labelbox) · 2026 – Present

- Authoring an engineering RL task for frontier AI models: problem design, reference solution and a deterministic Python grader, reviewed through a Git/CI workflow.

Graduate Teaching Assistant — IIT Kanpur · Jul 2023 – May 2024; Dec 2025 – Present

Teaching Faculty, Engineering Physics, Chemistry & Mathematics — GeeksforGurukul, Indore · Nov 2024 – May 2025

Doubt-Solving Educator (freelance, remote) — Unacademy · Aug 2021 – Jul 2022

- Recorded 200+ step-by-step video solutions to GATE Mechanical questions.

Education

PhD, Materials Science & Engineering (ongoing) — IIT Kanpur · Dec 2025 – present

Coursework: Thermodynamics of Materials; Fracture, Fatigue and Creep; AI & ML in Materials Engineering; Mathematics & Computational Methods. Ongoing: Transport Phenomena; Microscopy & Microanalysis.

M.Tech, Aerospace Engineering — IIT Kanpur · 2022 – 2024

B.Tech, Mechanical Engineering — Rajkiya Engineering College, Banda · 2018 – 2022

GATE 2021 (Mechanical), third year of B.Tech — AIR 8,664 of 1,20,594 (top 7.2%); GATE 2024 (Computer Science & IT) — AIR 22,651

Skills

- **Tools:** Quantum ESPRESSO, ASE, LAMMPS, phase-field modelling, ANSYS, SolidWorks, AutoCAD; Python (NumPy, SciPy, pandas, scikit-learn, pymatgen, pytest), MATLAB, C/C++, SQL, Git. **Languages:** English (professional), Hindi (native)